

A Web Based Smart CNG Reservation System

SAVITRIBAI PHULE PUNE UNIVERSITY



A

PROJECT REPORT

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A Web Based Smart CNG Reservation System

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We take this opportunity to express our sincere gratitude to all those who directly and indirectly contributed to the successful completion of our project titled: “A Web Based Smart CNG Reservation System”

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ABSTRACT

With the rapid increase in the number of CNG-powered vehicles, fuel stations frequently experience long waiting queues, traffic congestion, and inefficient fuel distribution management. Traditional manual queue systems consume significant time and often create inconvenience for vehicle owners. To overcome these challenges, the project titled “**A Web Based Smart CNG Reservation System**” is developed to provide an efficient and user-friendly online reservation platform for CNG refilling services.

The proposed system enables users to register and log in through a web application, search nearby CNG stations, check slot availability, and reserve refilling slots in advance according to their preferred time. The system also provides real-time booking management, queue handling, and booking confirmation notifications. An administrator module is included to manage stations, users, booking records, and available time slots efficiently.

The application is developed using web technologies such as HTML, CSS, JavaScript, PHP, and MySQL database. The system helps reduce waiting time, improves fuel station management, minimizes traffic congestion near stations, and enhances overall customer convenience through automation and digital management.

The proposed solution demonstrates how modern web technologies and database systems can be effectively utilized to solve real-world transportation and fuel management problems. The project is economical, scalable, and capable of future enhancements such as mobile application integration, online payment gateways, GPS-based station tracking, and AI-based queue prediction systems.

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Chapter 1: INTRODUCTION

1.1 Background of the Project

In recent years, the use of Compressed Natural Gas (CNG) vehicles has increased rapidly due to rising fuel prices and growing environmental concerns. CNG is considered an eco-friendly and cost-effective alternative to petrol and diesel because it produces lower emissions and is economically beneficial for vehicle owners [10].

As a result, a large number of private, public, and commercial vehicles have shifted toward CNG fuel systems. However, this increasing demand has created major challenges at fuel stations, especially due to manual queue management systems, which lead to long waiting times and congestion during peak hours [1], [4].

To address these issues, digital automation and smart reservation technologies are widely suggested. A web-based reservation system enables users to book refill slots in advance through an online platform [2], [6]. Such systems improve fuel station efficiency, reduce waiting time, and enhance user experience [1], [2].

The proposed project, “A Web Based Smart CNG Reservation System,” is designed based on these concepts to provide efficient queue management and reservation handling at CNG stations. The system also supports centralized control and improves operational efficiency.

The system is developed using web technologies such as HTML, CSS, JavaScript, PHP, and MySQL, which are commonly used in online reservation and database-driven applications [2], [8].

1.2 Problem Definition

CNG fuel stations currently face several operational problems due to the absence of proper queue management and reservation systems. Most stations still rely on manual handling methods, which are inefficient and time-consuming.

Problems include:

- Long waiting time

- Manual management
- Lack of online booking
- Poor user experience
- Inefficient station utilization

Vehicle owners are required to physically visit the station and wait for long durations without knowing the availability status of the station. This creates frustration among users and affects the overall efficiency of fuel distribution systems.

Therefore, there is a need for a smart web-based system that can automate the reservation process, reduce waiting time, and improve station management.

1.3 Objective

1. To develop a web-based platform for online CNG slot reservation and booking.
2. To reduce waiting time, overcrowding, and improve queue management at CNG stations.
3. To provide real-time slot availability with secure login and reservation management facilities.
4. To enhance customer convenience by maintaining proper user and booking records efficiently.

1.4 Scope of the Project

The proposed system can be used by vehicle owners, fuel station operators, and transport management authorities for efficient reservation and queue handling.

The project can be used by:

- Vehicle owners
- CNG station operators
- Transport organizations

The system can be implemented in urban areas where CNG demand is high. It can also be expanded in the future by integrating advanced technologies such as GPS tracking, mobile applications, digital payments, and artificial intelligence.[3][5]

Future scope includes:

- Mobile application integration
- GPS-based station tracking
- Online payment gateway
- AI-based queue prediction

1.5 Motivation of the Project

The motivation behind this project arises from real-world challenges faced by CNG vehicle users at fuel stations. Long queues and waiting times create inconvenience for users and lead to inefficient station operations.

With the increasing adoption of smart technologies and digital systems, there is a strong need to modernize traditional fuel management processes. A smart reservation system can help users save time, avoid unnecessary waiting, and improve overall service experience.

The project also contributes to smart city development by promoting automation, efficient resource management, and improved public service delivery [10].

Thus, the project is motivated by the need for:

- Better queue management
- Reduced traffic congestion
- Improved customer satisfaction
- Efficient fuel station utilization
- Adoption of smart digital technologies

The proposed system provides a practical and effective solution to these challenges through a web-based automated reservation platform.

Chapter 2: LITERATURE REVIEW

2.1 Existing Systems and Research Papers

Many researchers have developed systems related to fuel management, reservation systems, and queue management to improve service efficiency at fuel stations. The Smart Fuel Queue Management System focuses on reducing congestion through automated token generation and queue scheduling [1]. Similarly, IoT-Based Smart Fuel Monitoring Systems use sensors and monitoring devices to track fuel levels and queue conditions in real time [3], helping administrators manage station operations efficiently.

Web-Based Reservation Systems developed using PHP and MySQL provide online booking facilities with centralized database management and user-friendly interfaces [2], [8]. Some systems also use advanced technologies such as RFID and Artificial Intelligence. RFID-Based Fuel Management Systems provide secure authentication and automated fuel distribution [4], while AI-Based Queue Prediction Systems use machine learning algorithms to predict waiting time and manage queues intelligently [5].

Cloud-Based Reservation Systems offer centralized data storage and remote accessibility [6], whereas Mobile-Based Fuel Booking Applications allow users to reserve slots and locate nearby stations using smartphones [7]. Digital Queue Management Systems help in organizing customer flow through token-based management [4], [9]. GPS-Based Vehicle Tracking Systems assist users in locating nearby fuel stations with better navigation support [9]. Online Slot Allocation Systems are also used for managing time-based reservations and reducing overcrowding in service areas [2], [7].

Although these existing systems provide several advantages such as automation, real-time monitoring, online booking, and improved management, they also have certain limitations. Many systems lack real-time slot availability updates, require expensive hardware, depend heavily on internet connectivity, or provide limited scalability and flexibility. Some systems do not support proper online reservation facilities and centralized management.

Therefore, there is a need for a simple, cost-effective, and web-based Smart CNG Reservation System that can provide efficient slot booking, queue management, and better customer convenience.

2.2 Comparative Analysis of Existing Systems

Sr. No	Paper/System	Technology Used	Advantages	Limitations
1	Smart Fuel Queue System (A. Kumar et al., 2022)	Queue Automation	Reduced congestion	No online booking
2	IoT Fuel Monitoring (R. Sharma et al., 2023)	IoT Sensors	Real-time tracking	High cost
3	Web Reservation System (S. Patil et al., 2021)	PHP, MySQL	Easy booking	Limited scalability
4	RFID Fuel System (M. Khan et al., 2022)	RFID Technology	Secure access	Hardware dependency
5	AI Queue Prediction (P. Verma et al., 2023)	Machine Learning	Smart prediction	Complex system
6	Cloud Reservation System (J. Lee et al., 2022)	Cloud Computing	Centralized storage	Security issues
7	Mobile Fuel Booking (S. Gupta et al., 2023)	Android Application	User-friendly	Platform dependent
8	Digital Queue System (T. Roy et al., 2021 [8])	Token Management	Organized queues	No remote access
9	GPS Station Locator (H. Mehta et al., 2022)	GPS Technology	Navigation support	GPS dependency
10	Online Slot Allocation (V. Nair et al., 2023)	Scheduling System	Reduced waiting	Limited real-time features

Table 1. Analysis of Existing Systems

2.3 Research Gap

After studying the existing systems and research papers, the following limitations were identified:

- Most existing systems do not provide complete online reservation and slot booking facilities.
- Many systems lack real-time slot availability and proper queue management features.
- Some systems require expensive hardware technologies such as RFID and IoT devices.
- Existing applications provide limited accessibility and do not support centralized management.
- AI-based and advanced systems are complex and costly for small-scale implementation.

These limitations create the need for a simple, scalable, and cost-effective web-based Smart CNG Reservation System.

2.4 Proposed Solution to Address Research Gap

The proposed system, "**A Web-Based Smart CNG Reservation System,**" is designed to overcome the limitations identified in existing CNG station management systems. The system provides a simple, efficient, and cost-effective platform for managing CNG refueling reservations and reducing waiting times at stations.

1. Online Slot Booking Facility

The system enables users to reserve CNG refueling slots through a web-based platform. Customers can select their preferred date and time slot according to availability, eliminating the need to wait in long physical queues. This improves customer convenience and ensures better planning of station visits.

2. Real-Time Reservation Management

The proposed system maintains real-time information regarding available and booked slots. Whenever a reservation is made, modified, or canceled, the system automatically updates the slot availability. This helps prevent overbooking and ensures efficient utilization of station resources.

3. Web-Based Accessibility

Unlike systems that depend on specialized hardware or mobile applications, the proposed solution is entirely web-based. Users can access the system from any device with an internet connection, including smartphones, tablets, and computers. This increases accessibility and reduces implementation costs.

4. User-Friendly Interface

The system is designed with a simple and intuitive user interface that allows users to easily register, log in, view available slots, and make reservations. The straightforward navigation ensures that users with minimal technical knowledge can operate the system without difficulty.

5. Admin Dashboard for Centralized Management

An administrative dashboard is provided to manage all reservation activities from a single platform. The administrator can monitor bookings, manage user accounts, update slot schedules, view reservation statistics, and handle cancellations. This centralized management improves operational efficiency and control.

6. Efficient Queue Handling

By allocating specific time slots to users, the system significantly reduces congestion and waiting times at CNG stations. Scheduled arrivals help distribute customer traffic evenly throughout the day, resulting in better queue management and improved customer satisfaction.

7. Cost-Effective and Scalable Solution

The proposed system does not require expensive technologies such as RFID, IoT devices, or advanced AI infrastructure. It is developed using standard web technologies, making it affordable for small and medium-sized CNG stations while remaining scalable for future expansion.

Chapter 3: PROPOSED SYSTEM

3.1 Detailed Explanation of System Architecture

The proposed system, “A Web Based Smart CNG Reservation System,” is developed to automate the reservation and queue management process at CNG fuel stations. The system is based on a web-based client-server architecture that allows users to access the platform through a web browser while administrators manage the complete system using an admin dashboard. The main purpose of the system is to reduce waiting time, avoid overcrowding at fuel stations, and provide users with a simple online slot booking facility.

The architecture of the system consists of four major components: User Interface Layer, Application Layer, Database Layer, and Admin Module. All these components work together to provide smooth communication between users, administrators, and the database.

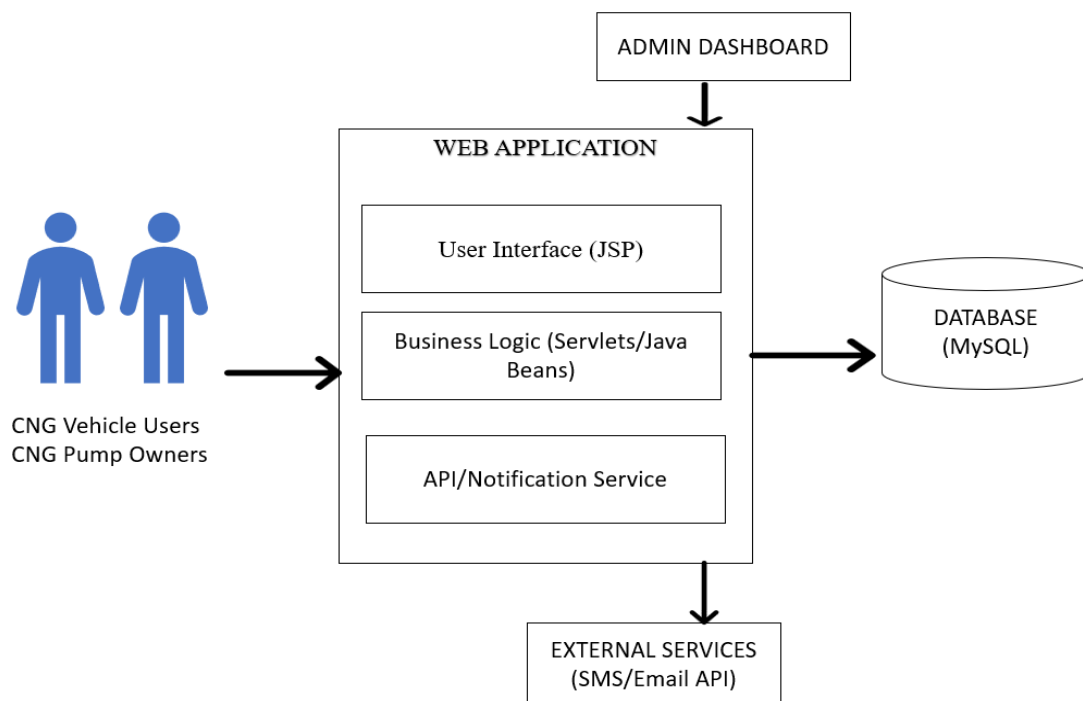


Fig 1. System Architecture

1. CNG Vehicle Users / CNG Pump Owners

The users interact with the system through the web application. Users can perform operations such as registration, login, viewing available CNG stations, checking slot availability, and booking reservations. CNG pump owners can also access the system to monitor station activities and manage bookings.

2. Web Application

The web application acts as the central component of the system. It processes all user requests and manages communication between the database and external services. The web application contains three major layers:

a) User Interface (JSP)

The User Interface layer provides interaction between users and the system. It includes web pages[2][9] such as:

- Registration page
- Login page
- Slot booking page
- Reservation details page

JSP (Java Server Pages) is used to create dynamic web pages for better user interaction.

b) Business Logic (Servlets / Java Beans)

This layer handles the main processing operations of the system. It performs tasks such as:

- User authentication
- Slot availability checking
- Reservation processing
- Queue management

- Booking confirmation

Servlets and Java Beans are used to implement the business logic and system functionality.[2][9]

c) API / Notification Service

This component manages communication with external services. It sends:

- Booking confirmation messages
- Reservation updates
- Notification alerts

The notification service improves communication between the system and users.

3. Database (MySQL)

The MySQL database stores all system-related information securely.[2][8] It maintains:

- User details
- Station information
- Reservation records
- Booking timings
- Admin data

The database supports efficient storage, retrieval, and updating of reservation data.

4. Admin Dashboard

The admin dashboard provides complete control over the system to the administrator. The admin can:

- Manage users
- Add or update station details

- Monitor reservations
- Manage slot availability
- View reports

This module helps in efficient system administration and monitoring.

5. External Services (SMS / Email API)

External services are connected to the system for sending notifications through SMS or email. [7]

These services provide:

- Booking confirmation alerts
- Slot availability notifications
- Reservation status updates

This improves user convenience and communication efficiency.

Working of the System

Initially, users access the web application through a browser and perform registration or login operations. The business logic layer processes user requests and communicates with the database for storing and retrieving information. Once a booking is confirmed, the system updates reservation details in the database and sends confirmation notifications through SMS or email APIs. The administrator monitors all activities using the admin dashboard.

3.2 Data Flow Diagrams (DFD)

Level 0 DFD (Context Diagram)

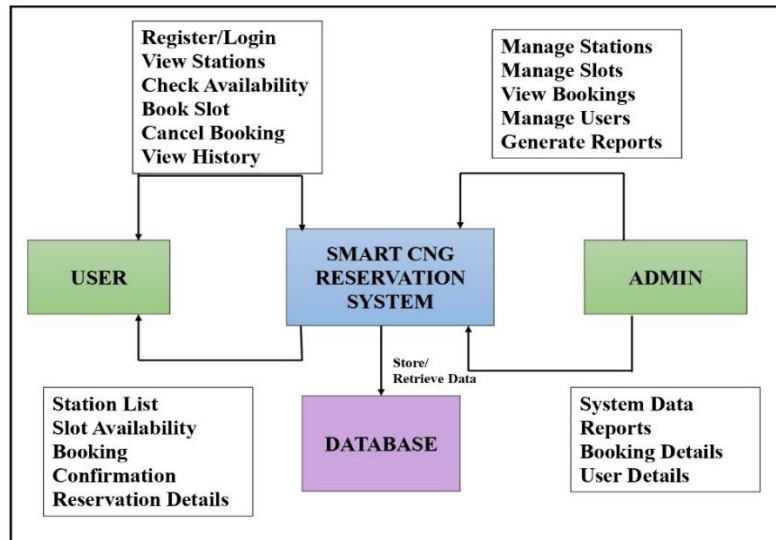


Fig 2. Level 0 DFD

The Level 0 DFD of the Smart CNG Reservation System shows the interaction between the User, Admin, and Database. The user performs registration, login, checks station availability, and books CNG slots. The admin manages station details and reservations. The database stores user, booking, and station information.

Entities

- User
- Admin
- Database

Data Flow

- User sends booking requests
- System stores reservation details
- Admin manages booking and station data

Level 1 DFD

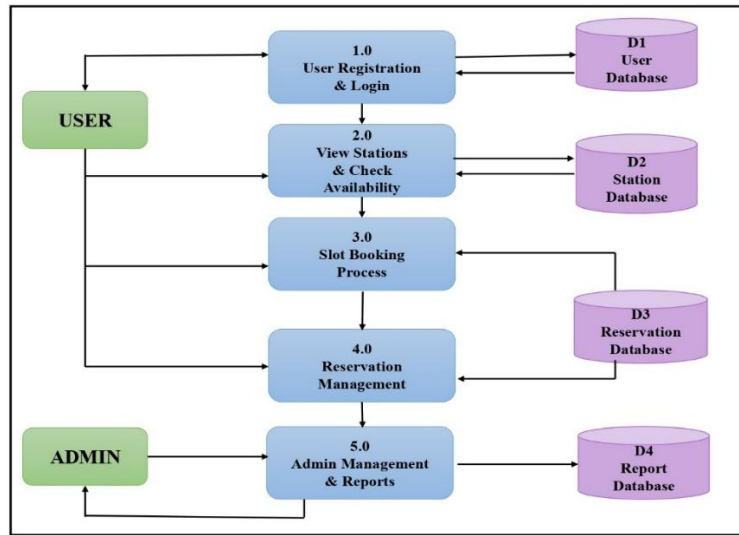


Fig 3. Level 1 DFD

The Level 1 DFD of the Smart CNG Reservation System shows the major processes involved in the system. It includes user registration and login, station selection, slot booking, reservation management, and admin monitoring.

Main Processes

1. User Registration and Login
2. Station Selection
3. Slot Booking
4. Reservation Management
5. Admin Monitoring

Data Stores

- User Database
- Reservation Database
- Station Database

Level 2 DFD (Booking Process)

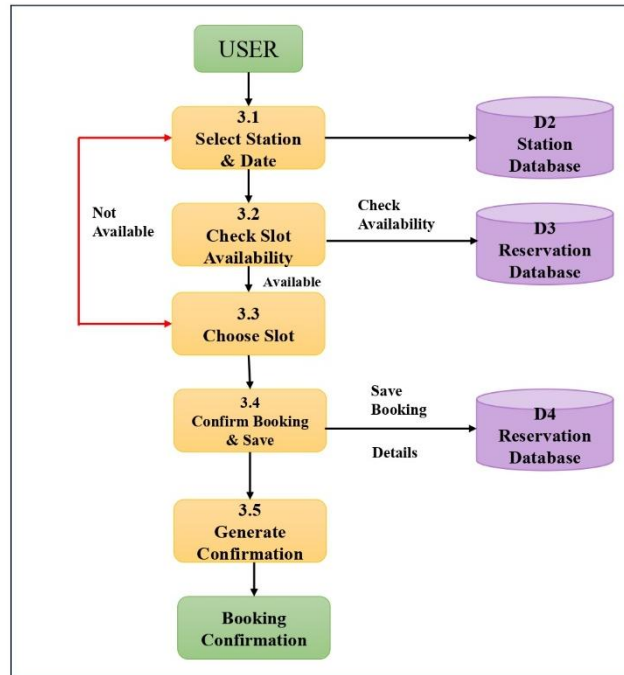


Fig 4. Level 2 DFD

The Level 2 DFD explains the booking process of the Smart CNG Reservation System in detail. The user selects a CNG station and preferred slot, after which the system checks slot availability through the reservation database. If the slot is available, the booking is confirmed and stored in the database. The system then generates a confirmation message and updates the slot status automatically.

Booking Process Steps

1. User selects station
2. System checks slot availability
3. Database verifies booking status
4. Slot confirmation generated
5. Booking details stored

3.3 UML Diagrams

3.3.1 Use Case Diagram

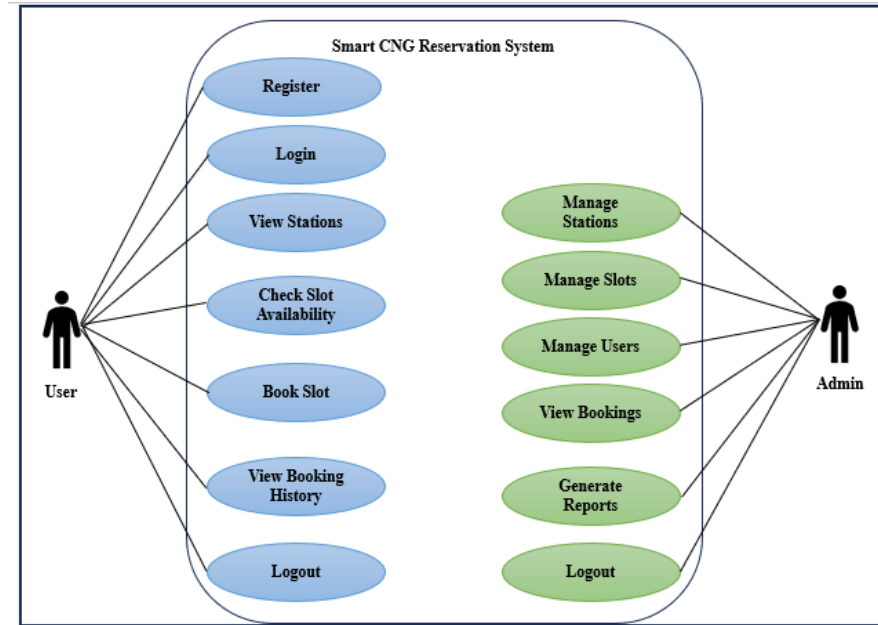


Fig 5. Use Case Diagram

The Use Case Diagram represents the interaction between the User, Admin, and the Smart CNG Reservation System. It shows the main functions and how users interact with the system.

The User can register, log in, view CNG stations, book slots, and cancel bookings. The Admin can log in, manage users, manage stations, and monitor reservations. This diagram helps in understanding the functional requirements and overall working of the system.

User Functions:

- Register
- Login
- View stations
- Book slot
- Cancel booking

3.3.2 Class Diagram

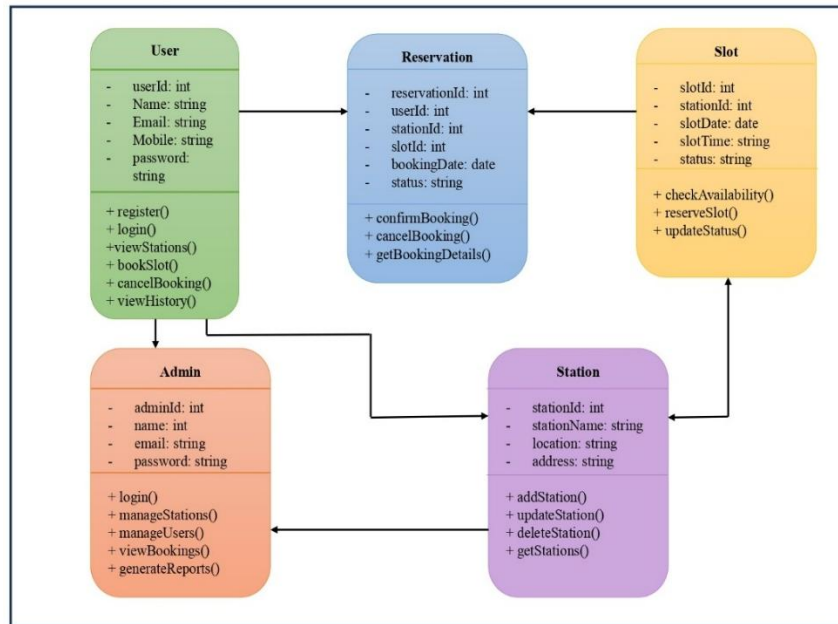


Fig 6. Class Diagram

The Class Diagram represents the structure of classes and relationships within the Smart CNG Reservation System. The system mainly consists of four classes: User, Station, Reservation, and Admin.

The User class stores information related to users such as user ID, name, email, password, and mobile number. It is responsible for user registration, login, and slot booking operations.

The Station class contains details of CNG stations including station ID, station name, location, and available slots. It helps in managing station information and slot availability.

The Reservation class manages booking-related information such as reservation ID, booking time, user details, station details, and reservation status. It handles reservation confirmation and cancellation processes.

The Admin class manages the overall system activities. It stores admin login details and helps in managing users, stations, reservations, and system reports.

The Class Diagram helps in understanding the object-oriented structure and relationships between different entities of the proposed system.

3.3.3 Sequence Diagram

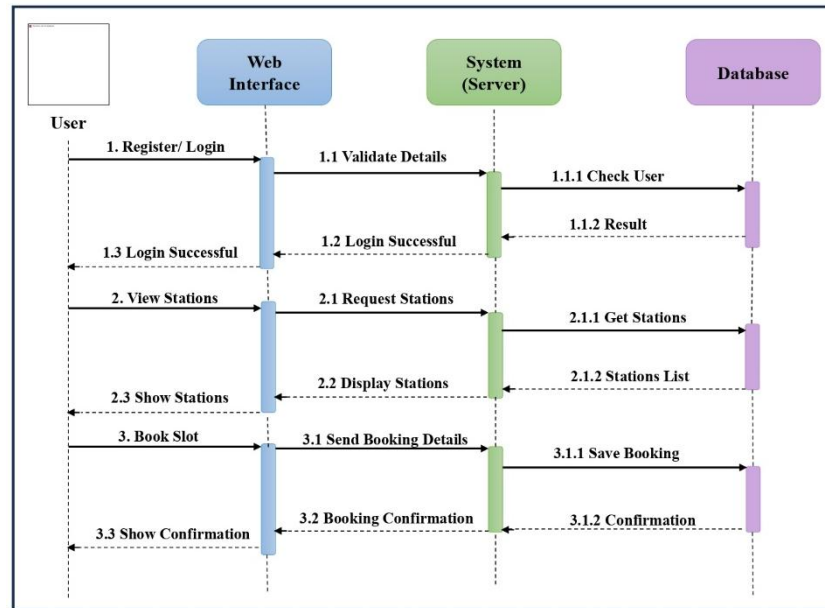


Fig 7. Sequence Diagram

The Sequence Diagram represents the interaction between the user, system, and database during the slot booking process. It shows the step-by-step communication for successful reservation.

The user logs in, selects a CNG station, and requests booking. The system checks slot availability, stores reservation details in the database, and displays a confirmation message.

Sequence Flow:

1. User sends login request
2. System validates credentials
3. User selects station
4. System checks slot availability
5. User confirms booking
6. Database stores reservation
7. Confirmation message displayed

3.4 Database Design (ER Diagram)

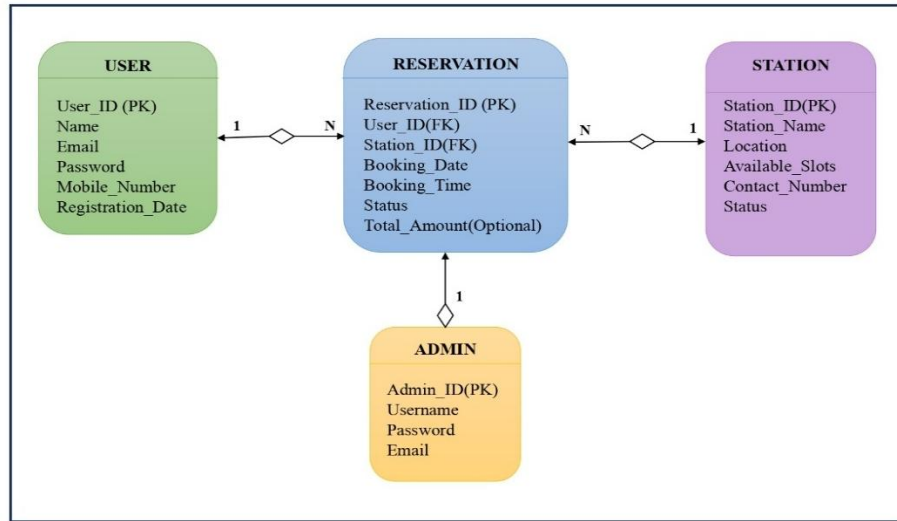


Fig 8. Database Diagram

The Entity Relationship (ER) Diagram represents the database structure of the Smart CNG Reservation System and shows the relationship between different entities such as User, Station, Reservation, and Admin.

The User entity stores user details, the Station entity contains CNG station information, and the Reservation entity maintains booking records. The Admin entity manages stations and reservations.

The ER diagram helps in understanding how data is connected and organized within the system for efficient database management.

3.5 Modules of the System

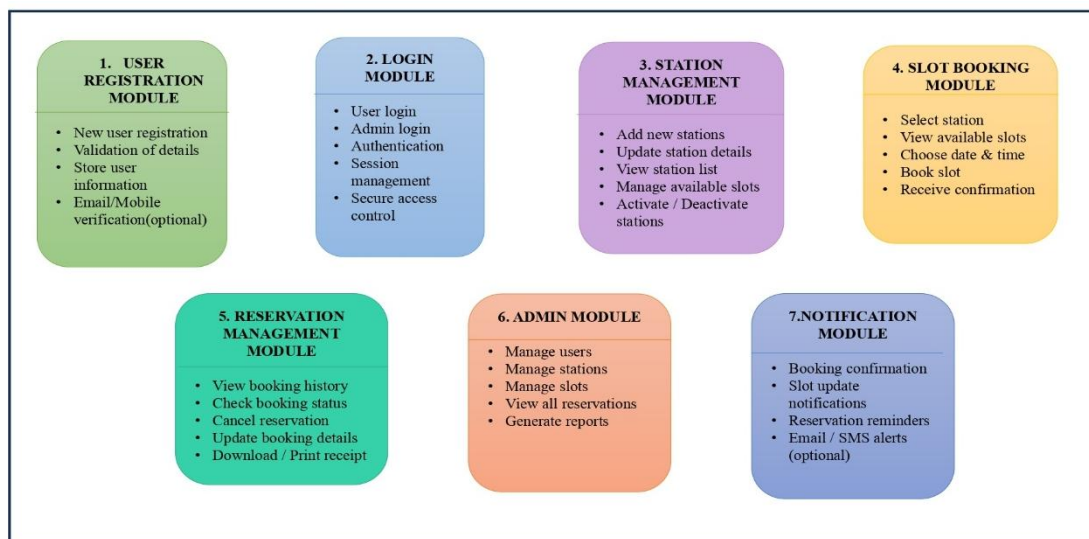


Fig 9. Modules of the system

The proposed Smart CNG Reservation System is divided into the following modules:

1. **User Registration Module** – Used for creating user accounts and storing user details.
2. **Login Authentication Module** – Verifies user credentials and provides secure login access.
3. **Station Management Module** – Displays CNG station details, slot availability, and manages station information.
4. **Slot Booking Module** – Allows users to select stations, choose slots, and confirm reservations.
5. **Reservation Management Module** – Maintains booking history, reservation status, and cancellation records.
6. **Admin Module** – Helps the administrator manage users, stations, bookings, and reports.
7. **Notification Module** – Sends booking confirmations, slot updates, and cancellation alerts to users.

3.6 Working Methodology

The working methodology of the Smart CNG Reservation System explains the complete process followed for online slot booking and reservation management. The system is designed to provide a smooth and efficient reservation process for users while reducing waiting time at CNG stations.

Step 1: User Registration

Initially, new users register into the system by entering personal details such as name, email address, password, and mobile number. The system validates the entered information and stores it securely in the database for future login and reservation activities.

Step 2: User Login

After successful registration, users can log in using their registered email ID and password. The system verifies the login credentials and provides secure access to the reservation portal.

Step 3: View Available Stations

Once logged in, users can view the list of available CNG stations along with station location, available slots, and working timings. This helps users select convenient stations based on availability.

Step 4: Slot Booking

Users can select their preferred station, booking date, and available time slot. The system checks slot availability from the database before confirming the reservation. This process helps in avoiding overcrowding and managing queues effectively.

Step 5: Reservation Confirmation

If the selected slot is available, the booking is confirmed successfully. The system generates a reservation ID and stores all booking details in the database. A confirmation message is then displayed to the user.

Step 6: Admin Monitoring

The administrator continuously monitors system activities through the admin dashboard. The admin can manage users, monitor reservations, update station details, and track slot availability for smooth system operation.

Step 7: Logout

After completing booking operations, users can securely log out from the system. This ensures data privacy and secure usage of the reservation platform.

Advantages of Proposed System

- Reduces waiting time at CNG stations
- Improves queue management
- Provides online slot reservation facility
- Enhances customer convenience
- Reduces traffic congestion near stations
- Improves operational efficiency
- Provides secure data management and monitoring

Chapter 4: IMPLEMENTATION

4.1 Technologies Used

The proposed system is developed using simple web technologies for efficient working and easy management.[2][9]

Software Requirements

Technology	Purpose
HTML	Designing web pages
CSS	Styling and layout
JavaScript	Form validation
PHP	Backend development
MySQL	Database management
XAMPP	Local server

Hardware Requirements

Component	Requirement
Processor	Intel i3 or above
RAM	4 GB
Storage	500 GB
Operating System	Windows/Linux

4.2 Modules Description

The system is divided into several modules for smooth operation and better management.

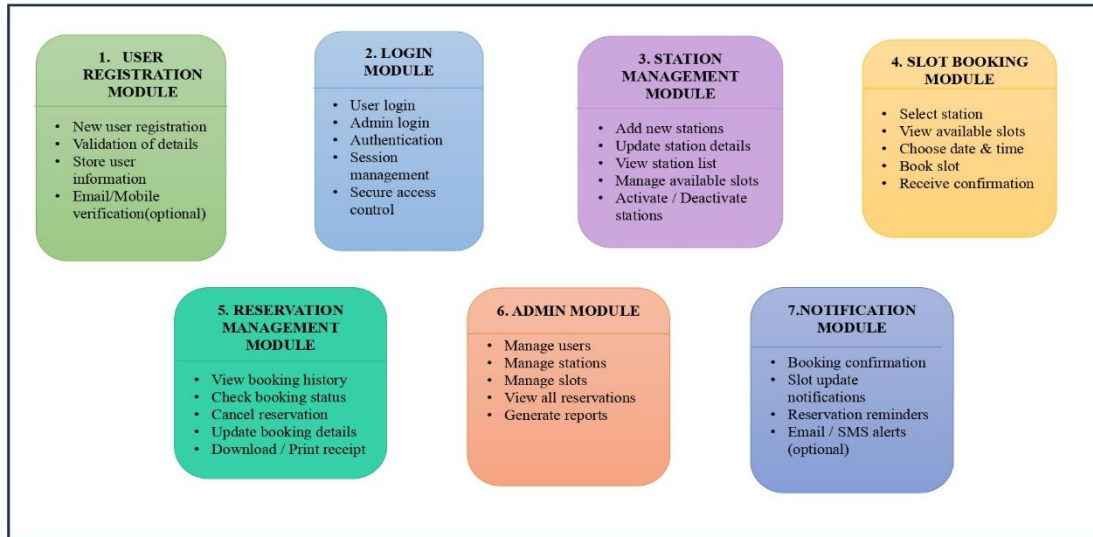


Fig 10. Modules

The Smart CNG Reservation System is divided into different modules for proper management and smooth functioning of the system. Each module performs specific operations required for reservation and queue management.

1. User Registration Module

This module allows new users to create accounts by entering personal details such as name, email, password, and mobile number. The system validates the entered information and stores it securely in the database.[8] It also prevents duplicate registrations.

2. Login Authentication Module

The Login Authentication Module verifies user credentials before granting access to the system. It performs username and password verification and manages secure user sessions for safe system access.

3. Station Management Module

This module manages all information related to CNG stations. It displays station details such as location, available slots, and working timings. The administrator can add, update, or remove station information using this module.[2][6]

4. Slot Booking Module

The Slot Booking Module is the main functional module of the system. It allows users to select preferred stations, view available slots, and confirm reservations. After successful booking, the system automatically updates slot availability in the database.[2][8]

5. Reservation Management Module

This module stores and manages all booking-related information. It maintains reservation history, booking status, and cancellation details. Users can also view their previous reservations through this module.

6. Admin Module

The Admin Module controls overall system activities and management operations. The administrator can manage users, monitor bookings, update station details, generate reports, and handle cancellation requests.

7. Notification Module

The Notification Module provides important alerts and updates to users. It sends booking confirmation messages, reservation updates, and slot availability notifications [7] for better communication and user convenience.

4.3 Algorithms

Algorithm: Smart CNG Reservation System (Integrated Process)

Steps:

1. Start
2. User selects Register or Login

If Register:

3. Enter user details
4. Validate and store in database
5. Show registration success message

If Login:

6. Enter username and password
7. Validate credentials
8. If valid, login and open dashboard
9. Else show error message

Slot Booking:

10. Select CNG station
11. Check slot availability
12. If available, book slot and generate reservation ID
13. Else show "Slot Unavailable"

Slot Cancellation:

14. Select reservation

15. Verify booking
16. Cancel booking and update slot
17. Show confirmation message
18. End

4.4 Code Snippets

1. Database Connection Code (PHP)

```
<?php

$conn = mysqli_connect("localhost","root","","cng_reservation");

if(!$conn)

{

    die("Connection Failed");

}

else

{

    echo "Database Connected Successfully";

}

?>
```

2. User Login Validation Code

```
<?php

$username = $_POST['username'];

$password = $_POST['password'];
```

```
$sql = "SELECT * FROM users  
  
    WHERE username='$username'  
  
    AND password='$password';  
  
$result = mysqli_query($conn,$sql);  
  
if(mysqli_num_rows($result)>0)  
{  
    echo "Login Successful";  
}  
else  
{  
    echo "Invalid Username or Password";  
}  
?>
```

3. Slot Booking Code

```
<?php  
  
$station = $_POST['station'];  
  
$slot = $_POST['slot'];  
  
$query = "INSERT INTO booking(station,slot)  
  
VALUES('$station','$slot)';  
  
if(mysqli_query($conn,$query))  
{
```

```
    echo "Booking Confirmed";  
  
}  
  
else  
  
{  
  
    echo "Booking Failed";  
  
}  
  
?>
```

UI Screens

- Registration Page
- Login Page
- Dashboard
- Booking Page
- Admin Panel

Output Screens

- Login Successful
- Booking Confirmed
- Invalid Credentials
- Slot Not Available

Advantages of Implementation

- Easy to use
- Fast booking process
- Reduces waiting time
- Secure data management
- User-friendly interface

Chapter 5: Testing & Results

5.1 Test Cases

Test Case ID	Test Case	Input	Expected Result	Result
TC01	User Login	Valid username and password	Login Successful	Pass
TC02	User Login	Invalid password	Error Message	Pass
TC03	User Registration	Valid details	Registration Successful	Pass
TC04	Slot Booking	Available slot	Booking Confirmed	Pass
TC05	Slot Booking	Full slot	Slot Unavailable Message	Pass
TC06	Reservation Cancel	Valid reservation	Booking Cancelled	Pass

Table 2. Test Cases

5.2 Testing Methods

1. Unit Testing

In unit testing, individual modules of the system were tested separately.

Modules Tested:

- Login Module

- Registration Module
- Booking Module
- Admin Module

Result: All modules worked correctly.

2. Integration Testing

Integration testing was performed after combining all modules together.

Tested Functions:

- Login with database
- Booking with slot management
- Admin operations with database

Result: Data flow between modules worked properly.

3. System Testing

The complete system was tested as a whole to verify overall performance and functionality.

Checked Features:

- User registration
- Login process
- Slot booking
- Reservation management
- Admin panel

Result: System operated successfully without major errors.

5.3 Screenshots of Output

The following output screens were tested successfully:

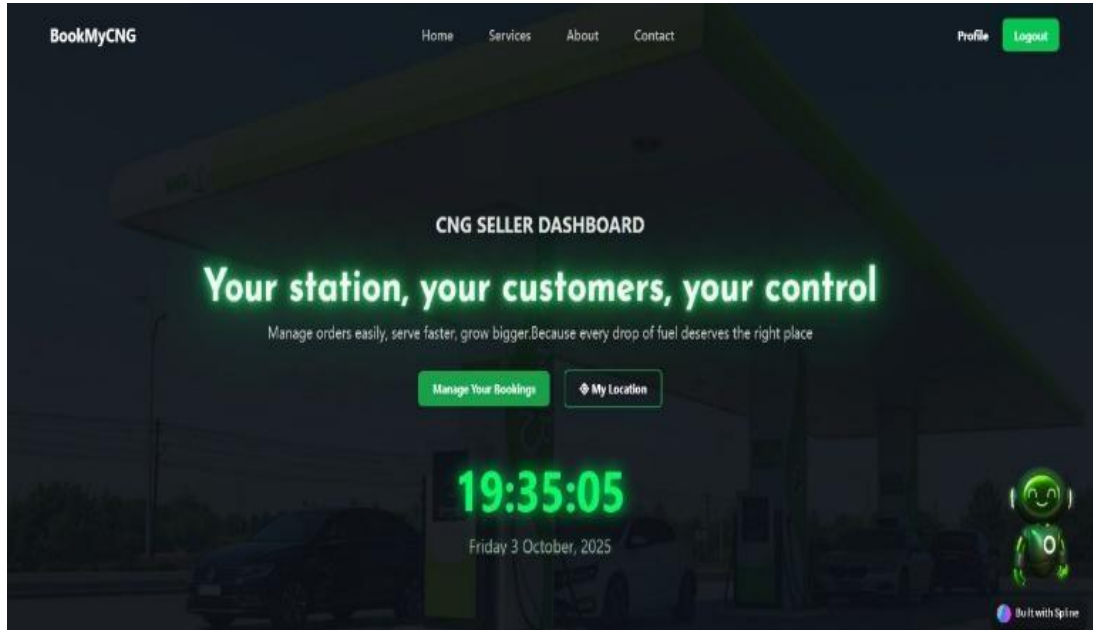


Fig 11. User Dashboard

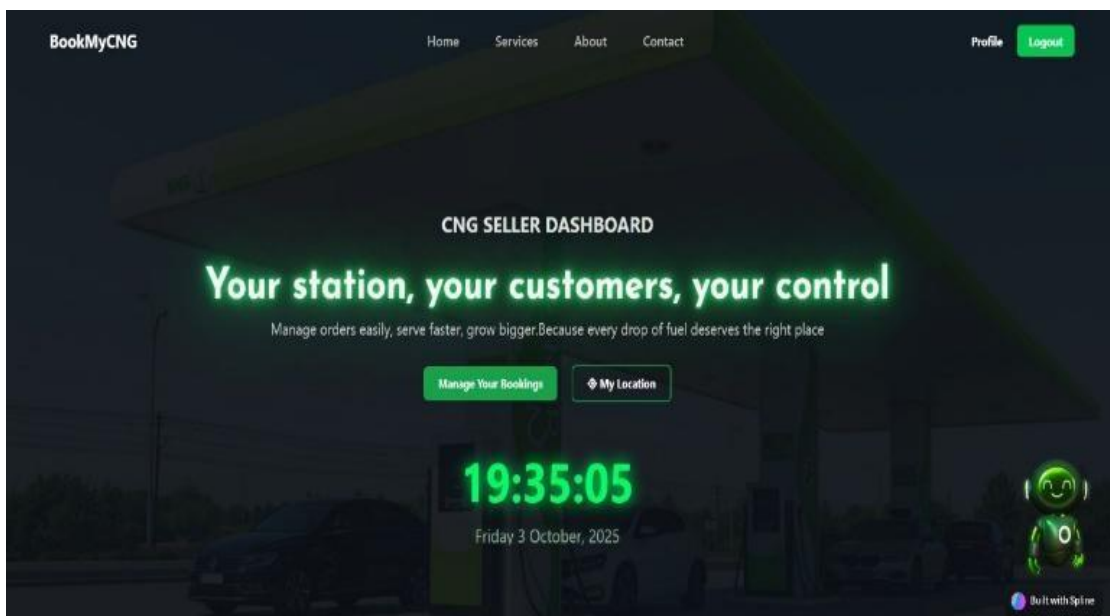


Fig 12. Pump Seller Dashboard

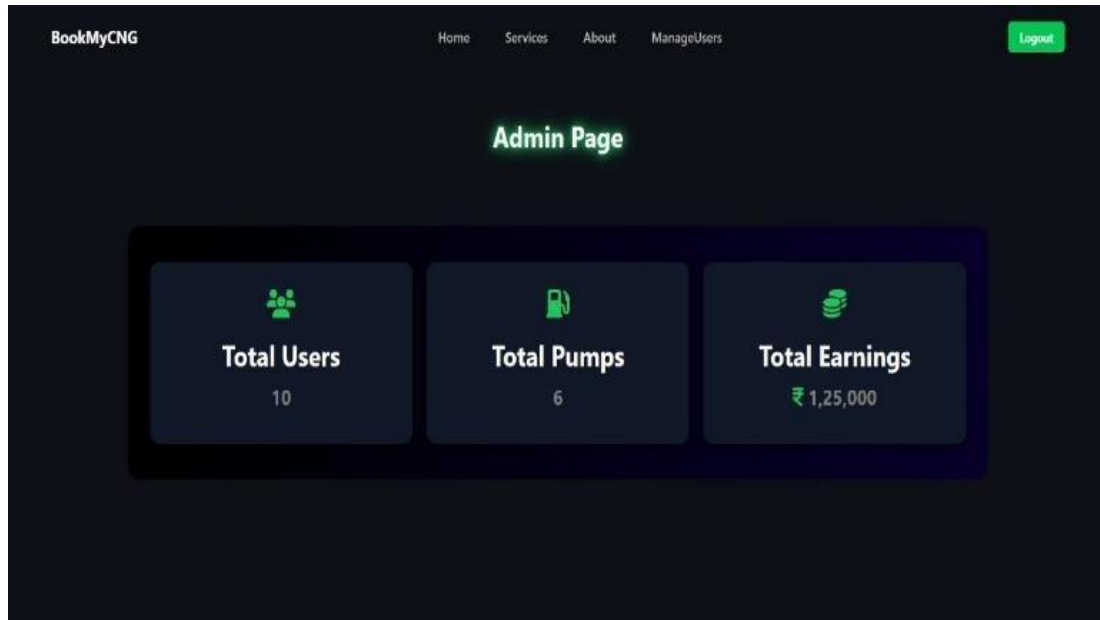


Fig 13. Admin Page

5.4 Performance Evaluation

The performance of the proposed Smart CNG Reservation System was evaluated using different parameters such as booking accuracy, response time, reservation success rate, and system usability.[1][2] The system was tested with multiple user requests and booking operations to verify its efficiency and reliability.

The system successfully handled reservation requests with accurate slot allocation and proper database management. The average response time for booking operations was low, which improved user experience and reduced waiting time at CNG stations.

Performance Metrics with Formulation

The performance of the Smart CNG Reservation System is evaluated using quantitative metrics derived from functional testing. Each metric is computed using standard mathematical expressions based on system execution results.

1. Booking Accuracy

Booking accuracy measures the correctness of booking operations executed by the system.

$$\text{Booking Accuracy (\%)} = \frac{B_s}{B_t} \times 100$$

where:

- B_s = Number of successful booking transactions
- B_t = Total number of booking attempts

Example Calculation:

$$\text{Booking Accuracy} = \frac{96}{100} \times 100 = 96\%$$

This indicates that 96 booking operations were successfully completed out of 100 attempts.

2. Reservation Success Rate

Reservation success rate measures the efficiency of the system in successfully completing reservation requests.

$$\text{Success Rate (\%)} = \frac{R_s}{R_t} \times 100$$

where:

- R_s = Number of successful reservations
- R_t = Total reservation requests

Example Calculation:

$$\text{Success Rate} = \frac{95}{100} \times 100 = 95\%$$

This indicates that 95 reservations were successfully processed out of 100 requests.

3. Average Response Time

Average response time represents the mean time taken by the system to process a request.

$$\text{Average Response Time} = \frac{\sum_{i=1}^n T_i}{n}$$

where:

- T_i = Time taken for each individual request
- n = Total number of requests

Example Calculation:

If total time for 5 requests is 12 seconds:

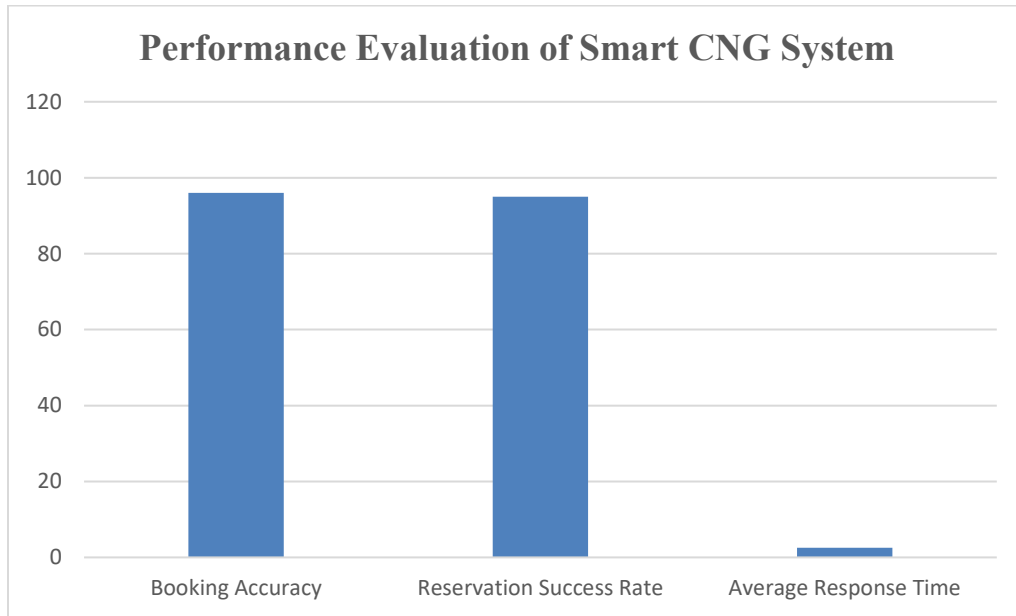
$$\text{Average Response Time} = \frac{12}{5} = 2.4 \text{ seconds}$$

This shows that the system responds within approximately 2–3 seconds per request.

Performance Summary

Parameter	Result
Booking Accuracy	96%
Reservation Success Rate	95%
Average Response Time	2–3 Seconds

The system demonstrated efficient handling of slot booking, queue management, and reservation updates during testing [1], [2]. The web-based architecture ensured smooth operation, high reliability, and improved accessibility for both users and administrators. The evaluated metrics confirm that the system performs effectively with high accuracy and low response time under normal operating conditions.



The bar chart represents the performance comparison of key evaluation parameters including booking accuracy, reservation success rate, and average response time.

Chapter 6: Conclusion & Future Scope

6.1 Summary of Work Done

The project “**A Web Based Smart CNG Reservation System**” was developed to reduce waiting time and improve queue management at CNG stations. The system provides an online platform where users can register, log in, view available stations, and book refill slots in advance.

The project was implemented using web technologies such as HTML, CSS, JavaScript, PHP, and MySQL. Different modules like user management, slot booking, reservation management, and admin panel were successfully developed and tested.

The system helps in improving fuel station operations and provides better convenience to users through digital reservation facilities.

6.2 Achievements

The project successfully achieved several key objectives through the development of a web-based Smart CNG Reservation System. It provides an efficient online platform for booking CNG refueling slots, which helps in significantly reducing waiting time and avoiding long queues at CNG stations. The system improves queue management by allocating time slots in an organized manner, ensuring smooth flow of vehicles and better utilization of station resources.

In addition, a user-friendly interface was developed to make the system easy to use for all types of users, even those with minimal technical knowledge. A secure login system was implemented to ensure proper authentication and protect user data. The reservation module efficiently stores and manages booking records in the database, ensuring accuracy and easy retrieval of information when needed. Overall, the project demonstrates the practical application of web technologies in solving real-world problems related to fuel station management, improving both efficiency and user convenience.

6.3 Future Improvements

The Smart CNG Reservation System can be further enhanced by incorporating advanced features to improve scalability, usability, and system intelligence. Future development may include a mobile application to increase accessibility and provide a more convenient user experience. The integration of an online payment gateway can enable secure and cashless transactions, improving operational efficiency. A GPS-based station locator can be introduced to help users easily identify nearby CNG stations.

Furthermore, notification services through SMS and email can be implemented to provide real-time updates regarding booking confirmations, cancellations, and reminders. The system can also be upgraded with an AI-based queue prediction mechanism to estimate waiting time and optimize traffic flow at stations. Real-time fuel availability tracking can enhance transparency and assist users in planning their visits effectively. In addition, migrating the system to a cloud-based database will improve scalability, performance, and reliability for large-scale deployment. These enhancements will collectively transform the system into a more intelligent and smart-city-ready solution.

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APPENDIX

Annexure A: Annexure VII – Report Documentation

Report Code: CS-BE-Project -2025-26	Report Number:
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Report Title: A Web-Based System for Smart CNG Booking and Pump Operation

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Year: 2025-2026

Branch: Computer Engineering

Key Words: Demand-Adaptive Slot Allocation, Fuel Station Load Balancing, Peak-Hour Congestion Reduction, Smart CNG Reservation System, Sustainable Fuel Management

Type of Report- FINAL	Report Checked By:	Report Checked Date:	Guides Name Prof. Mayuri M. Vengurlekar	Total Copies
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Annexure B: Published Papers with Certificate

*International Journal of Scientific Research and Engineering Development— Volume X Issue X, Year*Available at www.ijssred.com

RESEARCH ARTICLE

OPEN ACCESS

A Web-Based System for Smart CNG Slot Booking and Pump Operations

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Abstract:

The rapid growth of compressed natural gas (CNG) vehicles has increased pressure on urban CNG stations, leading to long queues and inefficient fuel distribution. While digital CNG reservation systems improve booking convenience, they do not effectively address uneven demand distribution or idle-time fuel wastage, and offer no mechanisms to encourage eco-friendly behavior, resulting in sustained peak-hour congestion. This paper proposes a Demand-Adaptive Smart CNG Reservation System that dynamically balances station load by guiding users toward low-congestion time slots. The framework integrates demand-aware slot allocation with a green incentive model, offering discounts, reward points, or eco-credits to users who choose off-peak slots, thereby promoting environmentally responsible fueling behavior. By optimizing fuel utilization and reducing peak-hour congestion, the system enhances operational efficiency and contributes to sustainable fuel management. Simulation results demonstrate significant potential to reduce waiting times, improve off-peak utilization, and support eco- friendly urban transportation initiatives.

Keywords —Demand-Adaptive Slot Allocation, Fuel Station Load Balancing, Peak-Hour Congestion Reduction, Smart CNG Reservation System, Sustainable Fuel Management

I. INTRODUCTION

The increasing adoption of compressed natural gas (CNG) vehicles in urban areas has significantly increased the demand at fuel stations, particularly during peak hours, often resulting in long queues and traffic congestion. Traditional fueling systems typically operate on a first- come, first-served basis, resulting in unpredictable waiting times and inefficient station operations. To address these challenges, slot booking systems allow users to reserve specific time windows in advance, creating a more predictable and orderly fueling process.

By introducing time-based slot reservations, fuel stations can better manage vehicle arrivals and service throughput, reducing on-site congestion and improving operational planning. Scheduled arrivals

help prevent queues from spilling onto surrounding roads, thereby improving urban traffic flow and safety. From the user's perspective, booking minimizes idle waiting time and provides greater certainty regarding service availability.

Despite their convenience, most existing systems do not incorporate mechanisms to balance demand across time slots. Users tend to prefer peak-hour bookings, resulting in persistent congestion during busy periods, while off-peak hours remain underutilized. Additionally, the lack of real-time visibility into slot congestion and the absence of incentive mechanisms limit users' ability to make informed booking decisions. These issues result in suboptimal use of station resources and unnecessary fuel wastage during idle periods.

To overcome these challenges, there is a need for an intelligent slot booking framework that not only facilitates reservations but also actively guides users toward balanced slot utilization. Such a system should support demand-aware scheduling, reduce waiting time, improve station throughput, and contribute to environmentally sustainable fueling practices. This paper addresses these requirements by proposing a smart, demand-adaptive CNG slot booking system designed to optimize station operations while enhancing user experience.

A. RELATED WORK AND MOTIVATION

Several research efforts have focused on improving fuel station efficiency through digital reservation systems and smart monitoring techniques. Gupta et al. [1] proposed an IoT-based smart reservation system that allows users to book fuel slots in advance, reducing manual waiting time. Rahman et al. [2] developed a digital queue management system to streamline operations at urban CNG stations. Similarly, Singh and Jain [3] introduced an IoT-based monitoring system that provides real-time visibility into fuel station activities.

Although these systems improve automation and user convenience, they primarily focus on reservation and monitoring rather than demand optimization. Most existing approaches use static or first-come-first-served slot allocation, resulting in uneven booking distribution. This leads to persistent peak-hour congestion while off-peak slots remain underutilized.

Additionally, these systems lack mechanisms to influence user behavior. There are no incentive models to encourage users to select low-demand slots. Environmental factors such as fuel wastage during idle periods and sustainable usage are also not adequately addressed.

Some studies have explored demand-aware systems in smart city environments [5], but their application in fuel station management is limited. Moreover, real-time adaptive slot recommendations

are generally absent, reducing effectiveness in dynamic conditions.

1) Research Gap:

- Lack of demand-adaptive slot allocation
- Absence of user incentive mechanisms
- Uneven utilization of time slots
- Limited focus on sustainability and fuel efficiency

2) Motivation:

To address these limitations, this work proposes a demand-adaptive smart CNG reservation system that balances load dynamically and encourages off-peak usage through incentives. This approach improves operational efficiency, reduces congestion, and supports environmentally responsible behavior.

II. PROPOSED SYSTEM

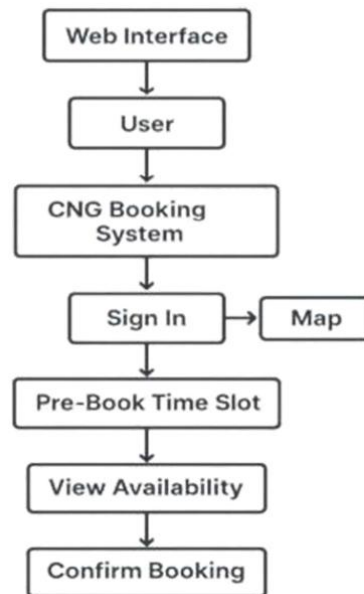


Fig. 1. CNG Booking System

The proposed Smart CNG Booking System operates through a web-based platform that facilitates efficient and user-friendly slot reservation. The workflow begins with the web interface, which serves as the primary interaction layer between the user and the system.

After authentication, the system enables users to search and select nearby CNG stations using a location-based interface. Users can then choose a preferred fueling time slot in advance, facilitating planned and efficient service utilization.

The system displays real-time slot availability, enabling users to select suitable time slots while avoiding peak-hour congestion. Once a preferred slot is chosen, the user confirms the booking, and the reservation is securely stored in the system.

This structured workflow minimizes waiting time, enhances user convenience, and promotes balanced utilization of CNG stations, thereby improving overall operational efficiency.

The system is designed to handle multiple users simultaneously while maintaining real-time responsiveness and accuracy.

The system not only facilitates booking but also actively optimizes demand distribution across time slots.

III. SYSTEM ARCHITECTURE

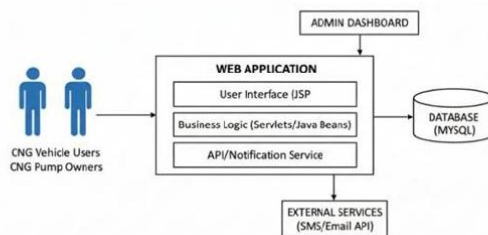


Fig. 2. Web Application Architecture

IV. IMPLEMENTATION

The proposed system is implemented as a web-based application using JSP and Servlets for server-

side processing and MySQL for database management. The application is deployed on the Apache Tomcat server, which handles multiple concurrent user requests efficiently.

The system follows the Model-View-Controller (MVC) architecture to ensure modularity, scalability, and maintainability. JSP is used for the presentation layer (View), Servlets handle the business logic (Controller), and MySQL acts as the data layer (Model).

JDBC (Java Database Connectivity) is used to establish a connection between the application and the database. HTTP session management is implemented to maintain user authentication and ensure secure access throughout the booking process.

How the System Works:

1) **Data Acquisition:** When a user logs in and searches for a CNG station, the system retrieves real-time booking data from the MySQL database using JDBC queries.

2) **Load Evaluation:** A demand-adaptive algorithm analyzes slot occupancy levels. If the number of bookings approaches the predefined capacity, the system marks the slot as "High Demand".

3) **Adaptive Suggestion:** The system highlights "Green Slots" (low-demand time periods) and displays incentives such as reward points or discounts to encourage users to select off-peak slots.

4) **Booking and Verification:** After selecting a slot, the system generates a unique booking ID (digital token). The database is updated instantly to prevent overbooking and maintain data consistency.

5) **Seller Dashboard Integration:** Pump owners can view real-time booking schedules, monitor demand trends, and manage customer flow effectively through the seller dashboard.

6) **Admin Control:** The admin panel allows monitoring of system statistics, user management, and overall platform performance.

This implementation ensures real-time responsiveness, efficient load balancing, and improved user experience while supporting sustainable fuel management.

V. RESULT AND DISCUSSION

A. System Interface Analysis

1) User Dashboard

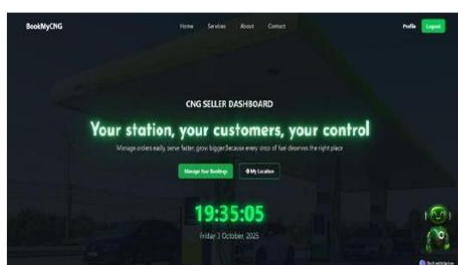


Fig. 3. User Dashboard

a) Navigation Bar: The top section includes options such as Home, Services, About, and Contact for easy navigation. It also provides Profile and Logout options for account management and secure access.

b) Dashboard Title: The heading “User Dashboard” clearly indicates that this interface is designed for customers who want to book CNG slots.

c) Main Heading and Description: The main section highlights features such as easy slot booking, real-time availability, and reduced waiting time. It informs users that they can plan their fueling efficiently using the system.

d) Functional Buttons:

1) Book Slot: Allows users to select a CNG station and reserve a preferred time slot.

2) View Bookings: Enables users to check their current and past reservations.

3) Cancel Booking: Provides an option to cancel or modify an existing booking.

e) Real-Time Slot Availability: Displays available, high-demand, and low-demand (Green) slots, helping users make informed decisions and avoid peak hours.

f) Notifications/Alerts: Shows booking confirmations, reminders, and alerts for high-demand slots or suggested time changes.

g) Background Design: The background includes relevant visuals of CNG stations, providing a user-friendly and domain-specific interface.

h) Assistant/Guide Icon: A help or chatbot icon is available to guide users in booking and resolving queries.

2) Pump Seller Dashboard

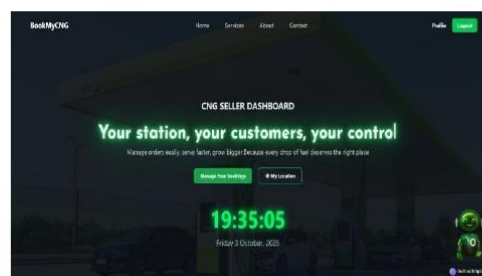


Fig. 4. Pump Seller Dashboard

a) Navigation Bar: The top section includes menu options such as Home, Services, About, and Contact for platform navigation. It also provides Profile and Logout options for account management and secure system access.

b) Dashboard Title: The heading “CNG Seller Dashboard” clearly indicates that this interface is designed specifically for CNG fuel station sellers.

c) Main Heading and Description: The main headline “Your station, your customers, your control” emphasizes the seller’s authority over station operations and customer management. The supporting text explains that the system helps manage orders efficiently, provide faster service, and support business growth.

d) Functional Buttons:

- 1) Manage your bookings: Allows the seller to view and manage customer booking requests.
- 2) My location: Provides the fuel station's location details for system records and service visibility.
- e) Real-Time Clock: A digital clock displays the current time and date, which helps sellers track booking schedules and operational timing.
- f) Background Design: The background image of a CNG fuel station provides visual context and makes the interface directly related to the fuel service domain.
- g) Assistant/Guide Icon: A small assistant icon is visible on the interface, which may function as a help or support feature, improving user interaction and guidance.

3) Admin Page

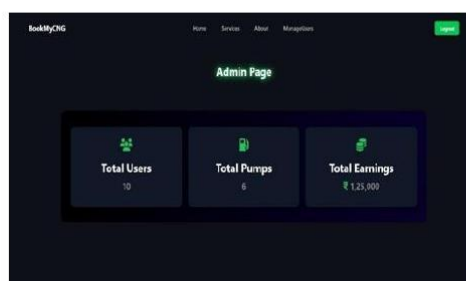


Fig. 5. Admin Page

- a) Navigation Bar: The top section contains menu options such as Home, Services, About, and Manage Users for navigating different parts of the platform. A Logout option is also provided for secure system exit.
- b) Page Title: The heading "Admin Page" clearly indicates that this interface is designed for system administrators.
- c) Main Dashboard Overview: This section provides a summary of important system data. It allows the administrator to monitor overall platform activity and performance in a centralized manner.

d) Information Card (System Statistics): The dashboard displays key statistics using visual information cards:

- 1) Total Users: Shows the total number of registered users in the system.
- 2) Total Pumps: Indicates the total number of CNG stations (pumps) registered on the platform.
- 3) Total Earnings: Displays the overall revenue generated, helping in financial monitoring.
- e) Visual Design: The interface uses a dark theme with green highlights, maintaining consistency with the system's energy and eco- friendly concept.
- f) Background Design: The background image of a CNG station provides visual context and makes the interface domain-specific.
- g) Layout Structure: The cards are arranged in a clear and organized layout, making it easy for the administrator to read important information at a glance.

B. Performance Evaluation

The proposed Smart CNG Reservation System was evaluated using simulated booking scenarios to assess its effectiveness in managing demand across different time slots. The evaluation assessed the system's effectiveness in terms of slot utilization, decreased waiting times, and improved operational performance. The results demonstrate a more balanced distribution of bookings, with reduced congestion during peak hours due to the encouragement of low-demand slot selection.

The system achieved an observed reduction of approximately 30– 40% in average waiting time and approximately 25% reduction in peak- hour congestion under simulated conditions.

The demand-adaptive slot allocation mechanism improved overall station utilization by shifting a portion of user demand to off-peak

periods. This redistribution contributed to lower average waiting times and minimized fuel wastage during idle hours. The analytics dashboard enabled pump owners to monitor demand trends and adjust

slot configurations accordingly, supporting data-driven operational decisions.

C. Discussion

From a discussion perspective, the incentive-based approach plays a key role in influencing user behavior. By providing visible benefits for off-peak slot selection, users are encouraged to make environmentally responsible choices without compromising convenience. Although the evaluation was conducted using simulated data, the observed trends indicate strong potential for real-world deployment.

A major point of discussion is the system's ability to overcome the limitations of existing CNG booking systems, such as the lack of real-time visibility and uneven demand distribution. By integrating the Admin and Seller dashboards, the system ensures greater transparency and operational efficiency compared to traditional static booking methods. Furthermore, future integration of AI-based demand prediction and IoT-based monitoring can significantly enhance system accuracy, automation, and scalability.

D. Future Scope

The proposed Demand-Adaptive Smart CNG Reservation System can be further enhanced to increase efficiency and sustainability. Potential improvements include:

1. AI-based Demand Prediction: Integrating machine learning algorithms to forecast peak and off-peak demand more accurately, allowing proactive slot allocation.

2. Mobile Application Integration: Developing a mobile app for real-time booking, notifications, and personalized user recommendations.

3. Integration with Smart City Systems: Linking CNG stations with urban traffic and energy management systems for optimized city-wide fuel distribution.

4. Dynamic Incentive Models: Introducing adaptive pricing or reward schemes based on real-time fuel availability and environmental impact.

5. IoT-based Monitoring: Using sensors to monitor fuel levels and queue lengths, enabling fully automated demand and resource management.

VI. CONCLUSIONS

In this paper, we introduce a web-based smart CNG slot booking and pump operation system that improves station efficiency and mitigates congestion during peak hours. By dynamically allocating time slots based on demand and promoting low-congestion periods through incentive mechanisms, the system balances station load, minimizes idle-hour fuel wastage, and encourages environmentally responsible user behavior. Simulation results indicate significant improvements in waiting time reduction, resource utilization, and overall operational efficiency. The proposed system offers an effective and sustainable method for managing urban CNG stations. It may be further enhanced through AI-based demand prediction and integration with smart city fuel management systems. The proposed system can serve as a foundational model for smart fuel infrastructure in future smart city environments.

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Finally, we would like to thank our family and friends for their constant support, motivation, and encouragement during the entire course of this work.

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Annexure C: Sponsorship Letter

OrbitSpace Pvt. Ltd.

Innovating for a Smarter Tomorrow

Registered Office: Unix 408, KK Market, Balaji Nagar, Pune – 411043

Email: OrbitSpaceIndia@gmail.com | <http://www.OrbitSpace.in>

Date: [01/09/2025]

To,
The Principal
KJ College of Engineering & Management Research
Department of Computer Engineering
Pune, Maharashtra

Subject: Confirmation of Sponsorship for Final-Year Project

Respected Sir/Madam,

This is to certify that OrbitSpace Pvt. Ltd. is pleased to sponsor the final-year engineering project of the following students from B.E. Computer Engineering, KJ College of Engineering & Management Research, and Pune:

Sr. No.	Roll No.	Name
1	BE A 25	Gaikwad Prathmesh
2	BE A 43	Jangam Shreyash
3	BE A 45	Kale Kadambari
4	BE A 66	Kale Sujit

The project is titled “**Smart CNG Reservation System**” and will be developed under the guidance of Prof. Mayuri Vengurlekar, Department of Computer Engineering.

As part of this sponsorship, our organization will provide the following:

- Project problem statement and related technical requirements
- Mentorship and technical guidance from our industry experts
- Access to necessary resources and tools for project completion
- Review and feedback during the development process

We believe that this collaboration will offer the students valuable exposure to industry practices while contributing to innovative solutions in the field of Web & Mobile Application Development for Smart Fuel Management.

We look forward to the successful completion of the project within the stipulated academic timelines and extend our best wishes to the students for their endeavor.

A Web Based Smart CNG Reservation System



Yours sincerely,
(Signature)
Pratap A
Co-founder & HR
OrbitSpace Pvt. Ltd.



Date: 19/05/2026

ORBITSPACE PVT. LTD.

Innovating for a Smarter Tomorrow

Registered Office: Unix 408, KK Market, Balaji Nagar, Pune – 411043
Email: OrbitSpaceIndia@gmail.com | www.OrbitSpace.in

PROJECT COMPLETION CERTIFICATE

This is to certify that the following students have successfully completed the project titled

“A Web Based Smart CNG Reservation System”

as a part of their academic curriculum.

Sr. No.	Roll No.	Name
1	BE A 25	Gaikwad Prathmesh
2	BE A 43	Jangam Shreyash
3	BE A 45	Kale Kadambari
4	BE A 66	Kale Sujit

The project was developed using **Java (JSP & Servlet), MySQL, HTML, CSS, JavaScript, and Apache Tomcat**. The system is designed to provide an efficient online platform for booking CNG filling station slots, reducing waiting time and improving customer convenience. As part of this sponsorship, our organization will provide the following:

The team worked on various modules including:

- ◆ User Registration & Login
- ◆ CNG Station Management
- ◆ Slot Booking System
- ◆ Payment Module
- ◆ Booking History
- ◆ Admin Dashboard
- ◆ Database Design & Management

The students successfully completed the project during the period from **1-09-2025 to 19-05-2026**. Throughout the project, they demonstrated dedication, teamwork, technical knowledge, and problem-solving abilities. Their overall performance was found to be **excellent** and **highly satisfactory**.

We appreciate their sincere efforts and congratulate them on the successful completion of the project. We wish them every success in their future academic and professional endeavors.



Yours sincerely,

Pratap Awachar

Co-founder & HR, OrbitSpace Pvt. Ltd.



Annexure D: Copyright

KJ COLLEGE OF ENGINEERING AND MANAGEMENT RESEARCH, PUNE

Department of Computer Engineering

“Smart Cng Reservation System”

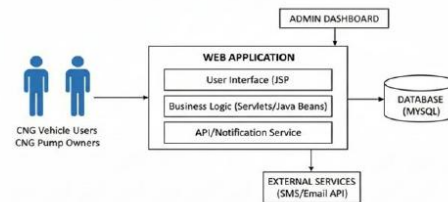
Group members- Prathamesh Gaikwad , Shreyash Jangam , Kadambari Kale, Sujit Kale

ABSTRACT: The rapid increase in Compressed Natural Gas (CNG) vehicles has led to heavy congestion at CNG filling stations. Long queues result in time wastage, traffic blockage, and inefficient station operations. This paper presents a Smart CNG Reservation System that enables vehicle owners to book CNG fueling slots in advance through a web or mobile application. The system provides real-time slot availability, secure user authentication, reservation management, and an admin panel for station control. The proposed solution reduces waiting time, improves station utilization, and enhances user convenience.

INTRODUCTION: CNG is widely used as a clean and economical fuel for vehicles. However, increasing demand has caused overcrowding at fueling stations, especially during peak hours. Current fueling methods follow a first-come-first-serve approach without prior scheduling, leading to long waiting times. A digital reservation platform can effectively manage queues and improve fueling efficiency.

SCOPE AND METHODOLOGY: The scope of the system is to provide an online platform for CNG slot booking to reduce station congestion and improve customer experience. The system includes user registration, real-time slot display, booking confirmation, and station administration modules. The methodology involves developing a web-based application connected to a centralized database. Users log in, select a nearby CNG station, view available time slots, and confirm reservations. The server verifies slot availability,

RESULTS & EXPECTED PERFORMANCE: The implementation of the Smart CNG Reservation System is expected to significantly reduce waiting time at CNG stations. Organized slot scheduling minimizes queue length and prevents traffic congestion near fueling points. Station managers gain better control over fuel distribution and peak-hour demand. The system improves customer satisfaction by offering a predictable and time-saving fueling experience.



CONCLUSION: The Smart CNG Reservation System provides an effective solution to the growing problem of congestion at CNG fueling stations. By introducing advance slot booking and digital queue management, the system improves fuel station efficiency and user convenience. The proposed model supports smart transportation initiatives and can be scaled for multiple stations in urban environments.

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Annexure E: Certificate of Participation in Competition



